Water Area

AlgoExpert · Hard · Dynamic Programming

The problem

- Pillars of given heights; how much rain water is trapped?
- Two pointers.

Intuition

- Water above a pillar is limited by the smaller of the tallest walls on its left and right.

Approach

- 1. Pointers at both ends with running maxes
- 2. Move the side with the smaller max
- 3. Add max minus height
- 4. Continue until they meet

Java solution

```java
public class WaterArea {
    public static int waterArea(int[] h) {
        int l = 0, r = h.length - 1, lMax = 0, rMax = 0, water = 0;
        while (l < r) {
            if (h[l] < h[r]) { lMax = Math.max(lMax, h[l]); water += lMax - h[l]; l++; }
            else { rMax = Math.max(rMax, h[r]); water += rMax - h[r]; r--; }
        }
        return water;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Water above a pillar is limited by the smaller of the tallest walls on its left and right.
- Complexity: O(n)